Where ED's gravity stands: passed, failed, open

Oct 19, 2026 · @Allen Proxmire · Runs 1–56 of the commitment-ontology lattice record

The short version

The lattice engine now matches every gravity test it has been set against, as far as it can resolve. The one failure found along the way, patterns bound inside a body falling wrongly (Run 45), was fixed by the full pattern rule. With matter in exact hydrostatic balance, adding energy to a star raises its far-field mass by that energy, as in GR, to within about 1% on the engine's grids; the old 4% shortfall is gone (Runs 50, 53, 56). In a medium with no built-in scale, bound matter falls exactly like its energy (Run 48). ED's medium does have a scale, so, taking a body's inertia to be its total energy (Allen's ruling), ED predicts that compact bodies slightly break the strong equivalence principle: a neutron star is pulled a little harder than its inertia (Runs 52–55). The pulsar triple bounds how much; ED does not yet fix the size.

- **Matched:** Newton, clocks slowing in gravity, everything falling alike, light bending, Mercury's orbit, the Earth–Moon test, frame dragging, binary-pulsar energy loss, bound energy pulling as mass, black-hole ringdowns and images.
- **How:** about a third come from ED's own rules. The rest build on established physics, as every new theory does: mostly "physics looks the same to a moving observer" (which GR also takes as a starting principle) and GR's black-hole exterior (which GR derives). They are kept as bookkeeping, marking where a later derivation could go deeper.
- **Failed:** many earlier ideas failed and were replaced. That is the record working as designed: each failure is kept, with what replaced it.
- **New predictions:** ED's central claim, that time runs one way at the bottom and the familiar time-symmetric physics is recovered by averaging, passed the engine's own registered test (Run 29). One new prediction already meets data, as a bound: compact bodies slightly break the strong equivalence principle, with a fixed sign (Run 55); the pulsar triple limits its size, which ED does not yet fix. ED's other claims (the arrow, a vacuum that does not pull, no singularities) have not yet been compared with data at physical scales.

Passed: from ED's own rules

Six results come straight from ED′s rules, with nothing chosen to match.

What is measured	What ED does	Runs
Gravity pulls as $1/r^2$ (Newton)	matter draws on a medium that holds possibility but never commits	1, 14
Clocks slow in gravity, all kinds alike	a commitment holds until enough possibility has passed	15, 22, 24
Everything falls alike	a pattern turned by the committed world′s count falls at the clock-set rate	16, 27
Mass deep in a well pulls by its local clock rate (GR′s Tolman term)	a body′s pull is its ticking	30
No preferred frame from possibility′s flow	follows from conservation, given the adopted rules	34
Time runs one way at the bottom; averages lose the order	commitments are never undone	29, 35, 37

Passed: building on established physics

Twelve results pass by building on established physics, as every theory does, or on a rule Allen chose. They are listed as bookkeeping: places where a later derivation, or a ruling that makes one a founding postulate, could go deeper.

What is measured	The input	Runs
Light bends; radar signals are delayed	patterns hop on their place′s ticks and need free room there (the pattern rule)	9, 17
Mercury′s orbit; two bodies′ wells multiply	the medium′s form, Candidate A′	22–26
Composite matter falls alike (MICROSCOPE)	physics looks the same to a moving observer (Lorentz invariance)	18b
Binary pulsars lose energy as GR says	relations between places can shear	11b

What is measured	The input	Runs
Different clocks agree; the Tolman bound; the pulsar triple	the medium′s two scales, p_s and P_0	23, 24, 30, 45, 55
Earth–Moon test, preferred-frame tests, frame dragging	Lorentz invariance at first order	34, 38
A bound system pulls by its total energy	a pattern′s energy draws (Allen) and its momentum flux draws (Lorentz input)	36, 40c
Compact bodies respond normally to outside gravity	bodies keep their proper size; for static bodies, now delivered by hydrostatic balance	40b, 46c
Binding inside a body pulls once	a body′s stress pulls (Allen)	42, 44, 50
Black-hole ringdowns and images	outside compact objects the field is GR′s (option A, Allen)	41, 43
Black holes stay stable, show no glow, leave no echoes	the core commits what reaches it (Allen)	43
Adding energy to a star raises its mass by exactly that energy (GR′s first law)	matter in hydrostatic balance (standard physics), in a medium with no built-in scale; ED′s rules then come from one energy principle	50

Failed, and what replaced it

Thirteen ideas failed a test and were replaced; each failure stays in the record.

What failed	Why	Replaced by	Runs
"Commitment wants to sync" as gravity	sync is lost between bodies at different depths	dropped as a gravity mechanism	2, 3, 7, 8
One cadence field radiating	gave 0.17 of GR′s radiated power	relations that shear	11

What failed	Why	Replaced by	Runs
The vacuum committing	it screened gravity	the vacuum never commits	14
ED′s original clock rule	no redshift where possibility is plentiful	the flow clock	15
Slowing possibility′s own hops	the medium froze	the pattern rule	17
Wells that add (β = ½)	Mercury at 50″ per century, not 43″	Candidate A′	19, 20
Candidate A′s clocks	clocks of different design disagreed	Candidate A′	21, 22
The engine′s own strong field	ringdown 4% off; spinning remnants fail GW250114	GR's exterior (option A)	28, 39, 41
Wells built by diffusion	too slow to follow a moving body	the field fixed each tick	31
No current field	Earth–Moon test failed	Lorentz input	34
Bodies held on fixed places	responded too weakly to outside gravity	bodies keep their proper size	36, 40
Attempt 03′s shallow core	would glow 25× brighter than Sgr A* allows	option A plus the core commits	43
Massive patterns with only the clock half of the pattern rule	bound energy fell with the wrong weight; as coded it would fail the pulsar triple about 100,000 times over	the full pattern rule (Allen); nothing fails down to the engine′s limit	45, 46–46c

Failed, not yet fixed

One failure is still open.

- **The strong-field mass bookkeeping (Run 40b).** In a very strong field, the engine misses a bound pattern's binding by about one binding. In ordinary gravity what is left is tiny (Run 44). With GR's exterior adopted for black holes, it matters only inside compact objects and possibly in neutron stars.

Open

Nine questions are open.

1. **Bound energy at neutron-star strength.** *Pulling:* with matter in exact point-by-point hydrostatic balance, ED's static rules come from a single energy principle, and adding a quantum raises a star's far-field mass by its energy. With four grids the registered test passes on the line: the continuum miss is within about 1% of the quantum's energy, and how the grid error shrinks is not settled (Runs 53, 56). The old shortfall was 4%. That is the same first law as GR's; it does not mean a neutron star's mass and radius match GR's, which is still open. *Falling:* in a medium with no built-in scale, bound matter falls exactly like its energy (Run 48). In ED's own medium a compact body is pulled slightly harder than its inertia, $\Delta \approx C \cdot p_s/P_0$, with $C \approx 0.33$–0.49 for the pulsar in the triple system (Run 55).
2. **What sets the medium's scale, P_0/p_s.** The draw rule that comes from one action (adopted 20 October) and inertia = total energy (ruled 20 October) turn ED's medium scale into a prediction: compact bodies slightly break the strong equivalence principle, $\Delta \approx C \cdot p_s/P_0$ (Run 55). ED does not yet fix P_0/p_s. If something in ED fixed it, the pulsar triple would become a sharp pass-or-fail test, with the sign already fixed ($\Delta > 0$).
3. **Neutron stars.** At first order, the pulsar triple tightens the medium's scale to $P_0/p_s \gtrsim 10^5$, about four times the old bound. A body's own binding passes its registered fall test (Run 46c, within 1 part in 1,000). At strong field, ED's own medium with inertia = energy gives $\Delta \approx C \cdot p_s/P_0$ for the pulsar, so the triple ($|\Delta| < 2.3$ in a million, or 1.5 with a planet in the timing model) needs $P_0/p_s \gtrsim 1.4$–2.9×10^5 (Run 55). Still open: a realistic neutron-star equation of state (the two tried differ by about 20%), and the pulsar's radius.
4. **Deriving the inputs:** Lorentz invariance (now extended to strongly self-gravitating bodies, so inertia = total energy), GR's black-hole exterior (option B, long-term), stress pulling, proper size.
5. **Signals through the instant field.** Shown to carry no signal only at first order.
6. **Black holes evaporate (Allen).** This rests on two assumptions: the core acts like a horizon for quantum fields, and its pull drops while its record stays. The endpoint is open: a Planck-mass remnant, or full evaporation (Allen leans this way).

7. **Does the radiation carry information out?** Paper 050 says yes, by two postulates. Mainstream physics has not settled this either.
8. **Superposed patterns.** Does a superposed pattern pull by its average energy, or only when it commits?
9. **The vacuum energy's size.** Its coefficient has not been computed.

Where ED says something new (not yet tested)

ED makes five claims GR and quantum mechanics do not. Only one has met data so far, and only as a bound (the last below).

- **Time's direction is fundamental.** This is ED's central claim, and it passed the engine's own registered test (Run 29): commitments are strictly one-way, and averaging over time recovers the time-symmetric-looking records that standard physics describes. What remains is comparing it with data at physical scales, where the leftover order (about c/T in a clock's record) is not yet known.
- **The vacuum never commits, so it does not pull.** That matches how small the observed cosmological constant is, but its size has not been computed.
- **No singularities.** Possibility per place is finite, so a well ends where it runs out (Paper 042).
- **Information is kept as record.** What falls into a black hole becomes part of the history, never lost. Whether the radiation carries it back out is not yet derived.
- **A small strong-equivalence-principle violation.** ED's medium has a built-in scale, so, with inertia = total energy, a neutron star is pulled slightly harder than its inertia: $\Delta \approx C \cdot p_s/P_0$, positive, growing with compactness, with $C \approx 0.33$–0.49 for the pulsar in PSR J0337+1715 (Run 55). GR gives exactly zero. The pulsar triple already bounds it ($P_0/p_s \gtrsim 1.4$–2.9×10^5); a measured nonzero value would fix ED's medium scale.

One surprising prediction that checks out would matter more than any number of matches.